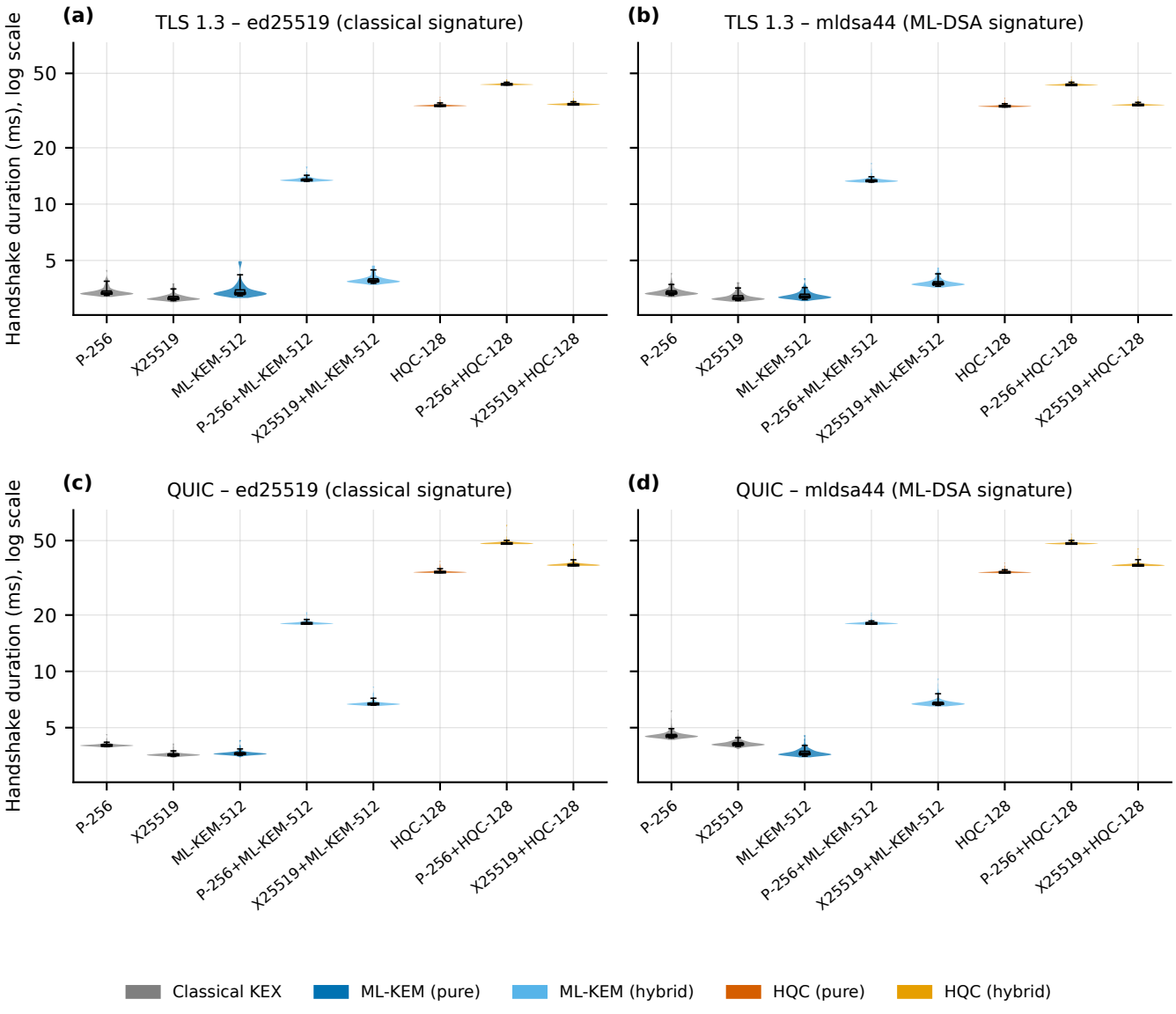

(a)
TLS 1.3 – ed25519 (classical signature)
(b)
TLS 1.3 – mldsa44 (ML-DSA signature)
(c)
QUIC – ed25519 (classical signature)
(d)
QUIC – mldsa44 (ML-DSA signature)
Handshake duration (ms), log scale
50
20
10
5
P-256
X25519
ML-KEM-512
P-256+ML-KEM-512
X25519+ML-KEM-512
HQC-128
P-256+HQC-128
X25519+HQC-128
Classical KEX
ML-KEM (pure)
ML-KEM (hybrid)
HQC (pure)
HQC (hybrid)